Forecasting and Macroeconomic Analytics

ECON 522

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Two Types of Breaks
Break in intercept (level shift): The mean of Y jumps up or down at some point. Example: COVID hits, GDP drops suddenly but then continues growing at roughly the same rate as before.
Break in slope (trend shift): The growth rate itself changes. Example: The “Great Moderation”—some economists argue GDP growth became more stable (lower volatility) after the mid-1980s, or that trend growth slowed after 2008.
You can have both simultaneously—a level shift and a change in the trend.

Structural Breaks

Breaks

Often we may want to model a break in the mean or a break in trend.

- We can use dummy variables to model that (time dummy variables can be used for many things...e.g. Covid)
- Break in the intercept or break in the slope of the trend? (think of an example)
- If we know the date of the break (new law, big event..) then we can impose that break. **We should test for significance of that break**.
- If we do not know the date of the break we can test for a break at unknown time.

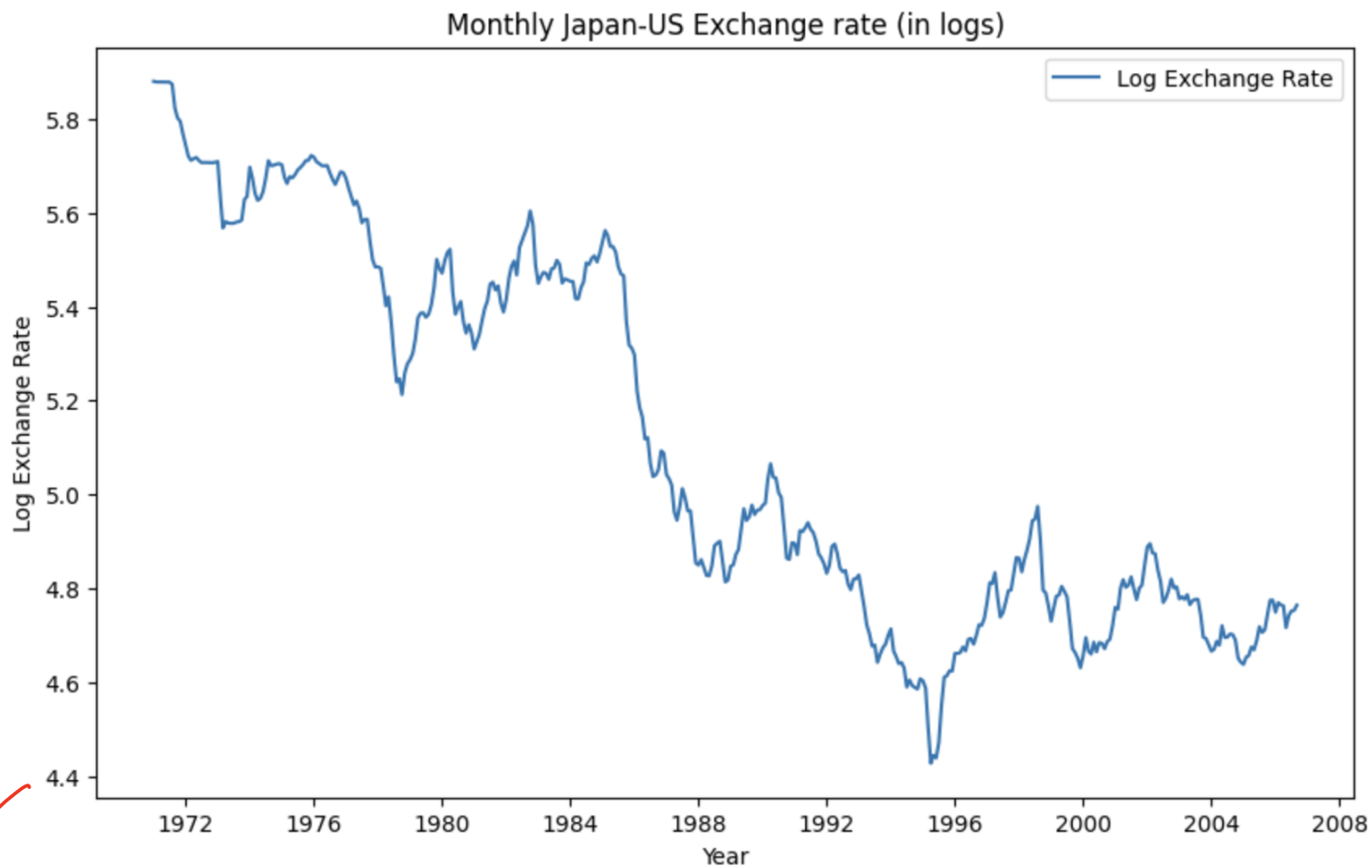
The Limitation

This model assumes the slope stayed the same—only the level shifted. Looking back at the exchange rate graph, that seems problematic. The pre-1986 period looked relatively flat, while the post-1986 period showed a clear downward trend.

A pure intercept break gives you two parallel lines, but the data suggests you might need non-parallel lines—which requires a slope break (interaction term $D_t \cdot t$) as well. k?

Broken Trend

Some time we may be interested in estimating a " broken trend".Often there are reason to expect that there is a structural break at a specific date.



Does this have a trend – tricky. Assume it does.

Trend tells you data was going down, and it will keep going down. This data ends at 2008, so you can put a trend, but you don't know what it looks like post 2008, it might go back up.

Why Joint Testing Matters

Problem 1: Multiple testing. If you do two separate tests at $\alpha = 0.05$, your actual Type I error rate is higher than 5%. You might reject "no break" just by chance on one of the tests.

Problem 2: The coefficients may be correlated. The estimates $\hat{\beta}_1$ and $\hat{\beta}_3$ aren't independent—they're estimated from the same regression, and the dummy and interaction term are correlated. A joint test (F-test) accounts for this covariance structure.

Problem 3: Different types of breaks. You could have:

- Level break only ($\beta_1 \neq 0$, $\beta_3 = 0$)
- Slope break only ($\beta_1 = 0$, $\beta_3 \neq 0$)

Exchange Rates example

impose

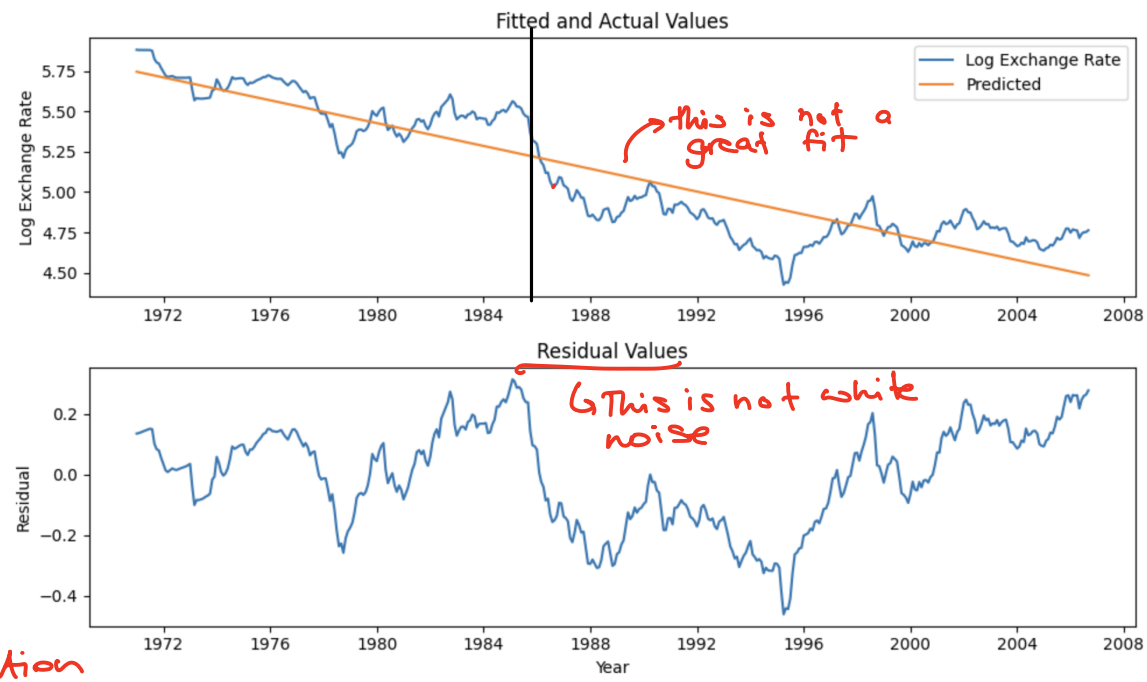
As before, we can of course estimate a linear trend

$$yen_t = c + \beta t + u_t$$

1986?

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                 Log_ER   R-squared:                       0.831
Model:                            OLS   Adj. R-squared:                  0.830
Method:                 Least Squares   F-statistic:                     2096.
Date:                Tue, 23 Jul 2024   Prob (F-statistic):          7.93e-167
Time:                        20:54:44   Log-Likelihood:                 165.73
No. Observations:                 429   AIC:                            -327.5
Df Residuals:                     427   BIC:                            -319.3
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Constant       5.7447      0.016    361.595      0.000       5.713       5.776
Time          -0.0029   6.43e-05    -45.786      0.000      -0.003      -0.003
==============================================================================
Omnibus:                       27.346   Durbin-Watson:                   0.027
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               17.741
Skew:                          -0.368   Prob(JB):                     0.000140
Kurtosis:                       2.328   Cond. No.                         494.
==============================================================================
```

Serial correlation



It looks like there is a break in the trend around 1986. Before mid 1986 the trend is flatter and it is decreasing after that!

Dummy Variables for breaks

To model this we use Dummy variables!

Your annotation is right—around 1986 something changed. This corresponds to the Plaza Accord (September 1985), where G5 nations agreed to depreciate the dollar against the yen and Deutsche Mark. That's a known policy intervention that fundamentally changed the exchange rate dynamics.
This is a perfect case for imposing a break at a known date. You'd want:

- A level dummy (did the intercept shift?)
- A slope dummy (did the trend change?)

Or possibly the relationship changed entirely and you need separate regressions for pre- and post-1986.
Is the next slide going to show the model with the break imposed?

Dummy Variables for breaks

To model this we use Dummy variables!
A **dummy variable** is a variables that only take the values of 0 and 1.
Define D_t as follows

$$D_t = \begin{cases} 0 \text{ if } t < \text{December 1985} \\ 1 \text{ if } t > \text{December 1985} \end{cases}$$

Let's see what model we are estimating if we include this variable in this regression.

$$yen_t = c + \beta_1 D_t + \beta_2 t + u_t$$

So you get two parallel lines—same slope (β_2), but different intercepts. The post-1986 line is shifted up or down by β_1.

testing for breaks in the Intercep

Clearly, before 1986, $D_t = 0$

$$yen_t = c + \beta_2 t + u_t$$

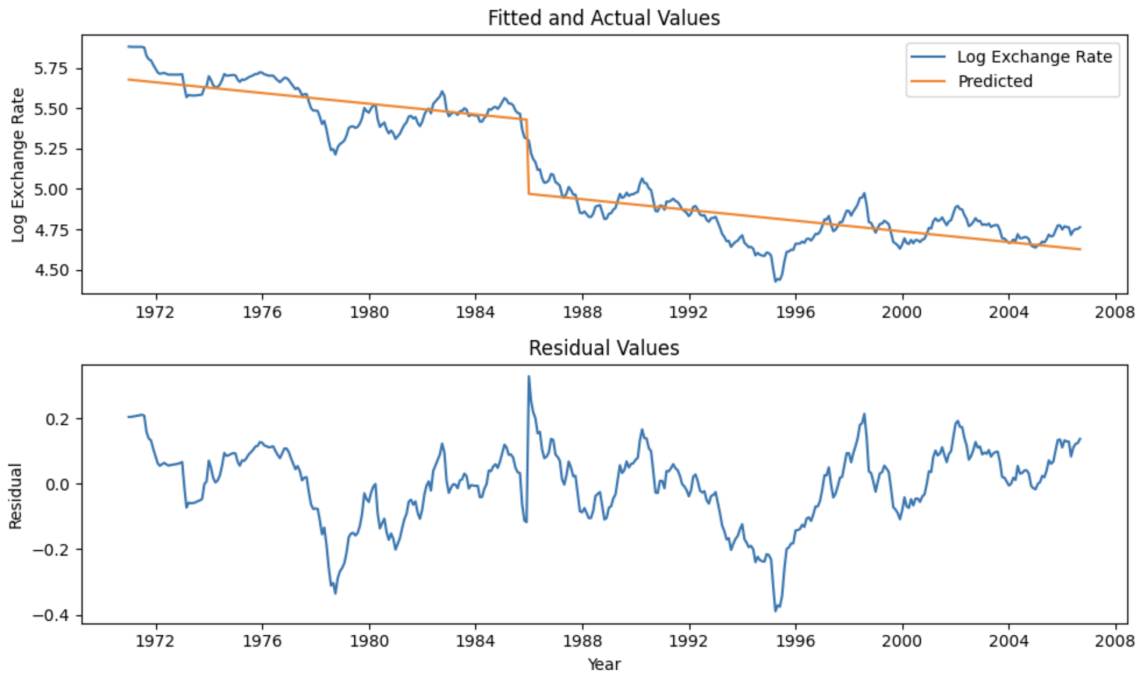
However, after 1986, $D_t = 1$

$$yen_t = (c + \beta_1) + \beta_2 t + u_t$$

Ergo, the coefficient on the dummy determines how the intercept has changed before and after 1986!

Testing Exchange rates

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                 Log_ER   R-squared:                       0.917
Model:                            OLS   Adj. R-squared:                  0.917
Method:                 Least Squares   F-statistic:                     2354.
Date:                Tue, 23 Jul 2024   Prob (F-statistic):          5.67e-231
Time:                        20:59:32   Log-Likelihood:                 318.55
No. Observations:                 429   AIC:                            -631.1
Df Residuals:                     426   BIC:                            -618.9
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Constant       5.6766      0.012    489.406      0.000       5.654       5.699
Post_85       -0.4583      0.022    -21.038      0.000      -0.501      -0.415
Time          -0.0014   8.68e-05    -15.912      0.000      -0.002      -0.001
==============================================================================
Omnibus:                       30.970   Durbin-Watson:                   0.091
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               35.794
Skew:                          -0.661   Prob(JB):                     1.69e-08
Kurtosis:                       3.507   Cond. No.                         979.
==============================================================================
```



- How many data points are used to estimate the mean in each time period?
- What does this tell us about how many data points we need in each period?

Breaks in the slope

We can use the same idea to allow for the possibility of a change in the slope. Use an "interaction" term $(D_t \cdot t)$

$$yen_t = c + \beta_1 D_t + \beta_2 t + \beta_3 (D_t \cdot t) + u_t$$

Before 1986, $D_t = 0$

$$yen_t = c + \beta_2 t + u_t$$

After 1986, $D_t = 1$

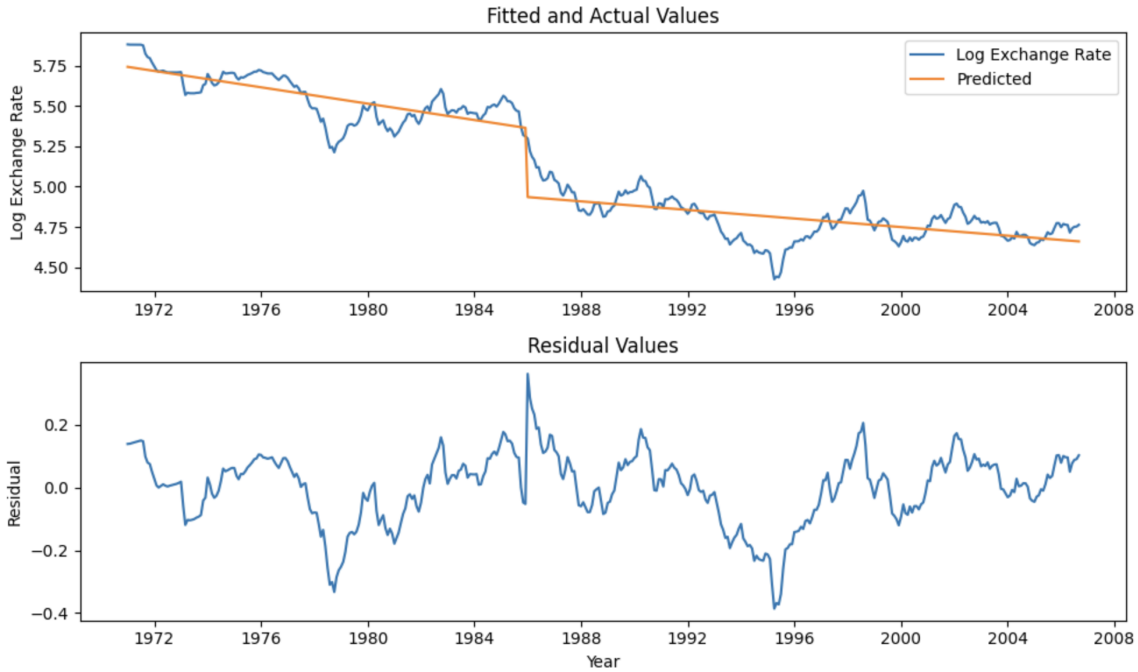
$$yen_t = (c + \beta_1) + (\beta_2 + \beta_3)\, t + u_t$$

The coefficient on the dummy determines how the intercept has changed before and after 1986, the coefficient on the interactive term determines how the slope has changed before and after 1986.

Testing Exchange rates for a break in the slope

slightly better than original model

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                 Log_ER   R-squared:                       0.922
Model:                            OLS   Adj. R-squared:                  0.922
Method:                 Least Squares   F-statistic:                     1679.
Date:                Tue, 23 Jul 2024   Prob (F-statistic):          3.64e-235
Time:                        21:03:39   Log-Likelihood:                 332.34
No. Observations:                 429   AIC:                            -656.7
Df Residuals:                     425   BIC:                            -640.4
Df Model:                           3
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Constant       5.7417      0.017    345.225      0.000       5.709       5.774
Post_85       -0.6069      0.035    -17.313      0.000      -0.676      -0.538
Time          -0.0021      0.000    -13.120      0.000      -0.002      -0.002
Time_Post_85   0.0010      0.000      5.312      0.000       0.001       0.001
==============================================================================
Omnibus:                       35.302   Durbin-Watson:                   0.092
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               43.676
Skew:                          -0.671   Prob(JB):                     3.28e-10
Kurtosis:                       3.800   Cond. No.                     2.27e+03
==============================================================================
```



Post-1986 slope = $\beta_2 + \beta_3$ = -0.0021 + 0.0010 = -0.0011

$yen_t = c + \beta_1 D_t + \beta_2 t + \beta_3 (D_t \cdot t) + u_t$

Before 1986: $yen_t = c + \beta_2 t$ (intercept c, slope β_2)

Need to do joint tests – is there a break?

$$\begin{cases} H_0: \beta_1 = \beta_3 = 0 \\ H_a: \text{at least one} \neq 0 \end{cases}$$

F-Test: / Chow Test

The Problem with Separate t-tests

Looking at the output, you might be tempted to test:

- Is β_1 (Post_85) significant? → t-test
- Is β_3 (Time_Post_85) significant? → t-test

But "is there a break?" is really asking whether both coefficients are jointly zero:

H_0: $\beta_1 = 0$ AND $\beta_3 = 0$ (no break at all)

H_1: $\beta_1 \neq 0$ OR $\beta_3 \neq 0$ (some kind of break exists)

Why Joint Testing Matters

Problem 1: Multiple testing. If you do two separate tests at $\alpha = 0.05$, your actual Type I error rate is higher than 5%. You might reject "no break" just by chance on one of the tests.

Problem 2: The coefficients may be correlated. The estimates $\hat{\beta}_1$ and $\hat{\beta}_3$ aren't independent—they're estimated from the same regression, and the dummy and interaction term are correlated. A joint test (F-test) accounts for this covariance structure.

Problem 3: Different types of breaks. You could have:

- Level break only ($\beta_1 \neq 0$, $\beta_3 = 0$)
- Slope break only ($\beta_1 = 0$, $\beta_3 \neq 0$)
- Both ($\beta_1 \neq 0$, $\beta_3 \neq 0$)

Individual t-tests might both be insignificant, but jointly they could still indicate a break—or vice versa.

Testing for Breaks

Why is important to detect breaks?

- If a break occurs in the population regression model in the sample, then the OLS will estimate a relationship that holds 'on average'. The average combines the two periods!
  - Depending on the location and the size of the break the estimated 'average' relationship can be very different from the true population line.
  - For this reason it is important that we test the presence of breaks in time series.

The Test

You'd run an F-test comparing:

- Restricted model: $yen_t = c + \beta_2 t + u_t$ (no break)
- Unrestricted model: $yen_t = c + \beta_1 D_t + \beta_2 t + \beta_3 (D_t \cdot t) + u_t$

The F-statistic tests whether the added terms jointly improve the model significantly.

Breaks at Known Time

- Sometime we may suspect that there is a break at a certain date (e.g. Bretton Woods (1973)).
- We showed how to use dummy variables to estimate possible breaks in the intercept and in the slope.
- If there is no break, then the regression line should be there same in both periods.

$$y_t = c + \beta_1 D_t + \beta_2 t + \beta_3 (D_t \cdot t) + u_t$$

- We want to test the null hypothesis of no break. What is the null? Which test can we use? **Answer: Let's try the Chow's Test!**

The Intuition

Instead of asking "are the break dummies significant?", the Chow test asks: would we do better

running two completely separate regressions?

You're comparing:

- Pooled model: One regression using all data, assuming no break
  - Split model: Two separate regressions, one for each sub-period

If the relationship is truly different before and after 1986, the split model should fit much better.

The Mechanics

Step 1: Run the pooled regression (all data), get RSS_pooled with n observations

Step 2: Run regression on pre-break data only, get RSS_1 with n_1 observations

Step 3: Run regression on post-break data only, get RSS_2 with n_2 observations

Step 4: Compute the F-statistic:

$F = [(RSS_pooled - (RSS_1 + RSS_2)) / k] / [(RSS_1 + RSS_2) / (n - 2k)]$

Where k is the number of parameters in each regression.

What It's Testing

H_0: The coefficients are the same in both periods (no break)

H_1: At least one coefficient differs across periods

This is equivalent to jointly testing whether the break dummies are significant, but the Chow framing makes the intuition clearer—you're asking whether forcing a single set of coefficients on both periods significantly worsens the fit.

Important limitation

The Chow test requires you to know the break date in advance. If you searched for the break date by looking at the data (as we did visually), the standard F critical values aren't valid—you'd need something like the Quandt-Andrews test instead

Operationalizing the Chow's Test

Consider the model

$$yen_t = c + \beta_1 D_t + \beta_2 t + \beta_3 D_t \cdot t + u_t$$

We want to test the hypothesis

$$H_0 : \beta_2 = \beta_3 = 0$$

$$H_a : \text{otherwise}$$

The null hypothesis of joint insignificance of D can be run as an F-test with k and $N_1 + N_2 - 2k$ degrees of freedom where N_1 is $card(t_1, ..., t^*)$ and N_2 is $card(t^* + 1, ..., T)$ and k parameters.

not expected to memorize

$$F = \frac{(RSS_C - (RSS_1 + RSS_2))/k}{(RSS1 + RSS_2)/(N_1 + N_2 - 2k)}$$

Note: The same result can be achieved with dummy variables!

Operationalizing the Chow's Test

```python
# Run a Chow test to test for structural break

# Define the two subsamples
er_pre = er[er['DATE'] <= '1985-12-31']
er_post = er[er['DATE'] > '1985-12-31']

# Run the regression for the two subsamples
model_pre = sm.OLS(er_pre['Log_ER'], er_pre[['Constant', 'Time']])
results_pre = model_pre.fit()

model_post = sm.OLS(er_post['Log_ER'], er_post[['Constant', 'Time']])
results_post = model_post.fit()

# Compute the sum of squared residuals for the two subsamples
SSR_pre = np.sum(results_pre.resid**2)
SSR_post = np.sum(results_post.resid**2)

# Compute the total sum of squared residuals
SSR_total = np.sum((er['Log_ER'] - np.mean(er['Log_ER']))**2)

# Compute the Chow test statistic
Chow = ((SSR_total - (SSR_pre + SSR_post)) / 2) / ((SSR_pre + SSR_post) / (len(er) - 4))
Chow
```

[17] ✓ 0.0s Python

```
2518.029081609119
```

Clearly, we find that we reject the null hypothesis! The same conclusion was reached through the use of dummy variables.

Breaks at Unknown Time

- More often we don't know when the break occurred but we suspect that it was sometime between date t_0 and t_1. This is a much more interesting question!
- The Chow test can be modified to to handle this by testing for breaks at all possible dates between t_0 and t_1 and then taking the largest of the resulting F-tests.
- This is called the **Quandt test** (or sup-Wald test)
- Because the Quandt test is the *largest* of an individual F-test, its distribution is not the same as the individual tests but it will have its own distribution.

This is the more realistic scenario—you suspect there's a break somewhere, but you don't know exactly when.

The Quandt Test (Sup-Wald Test) Logic

Step 1: Define a window $[t_0, t_1]$ where you think the break might have occurred. You typically trim the endpoints (e.g., exclude the first and last 15% of observations) because you need enough data on both sides to estimate the regressions.

Step 2: Run a Chow test at every possible break date within that window. So if your window has 100 possible break points, you get 100 F-statistics.

Step 3: Take the maximum F-statistic across all those tests. This is your Quandt statistic (also called sup-F or sup-Wald).

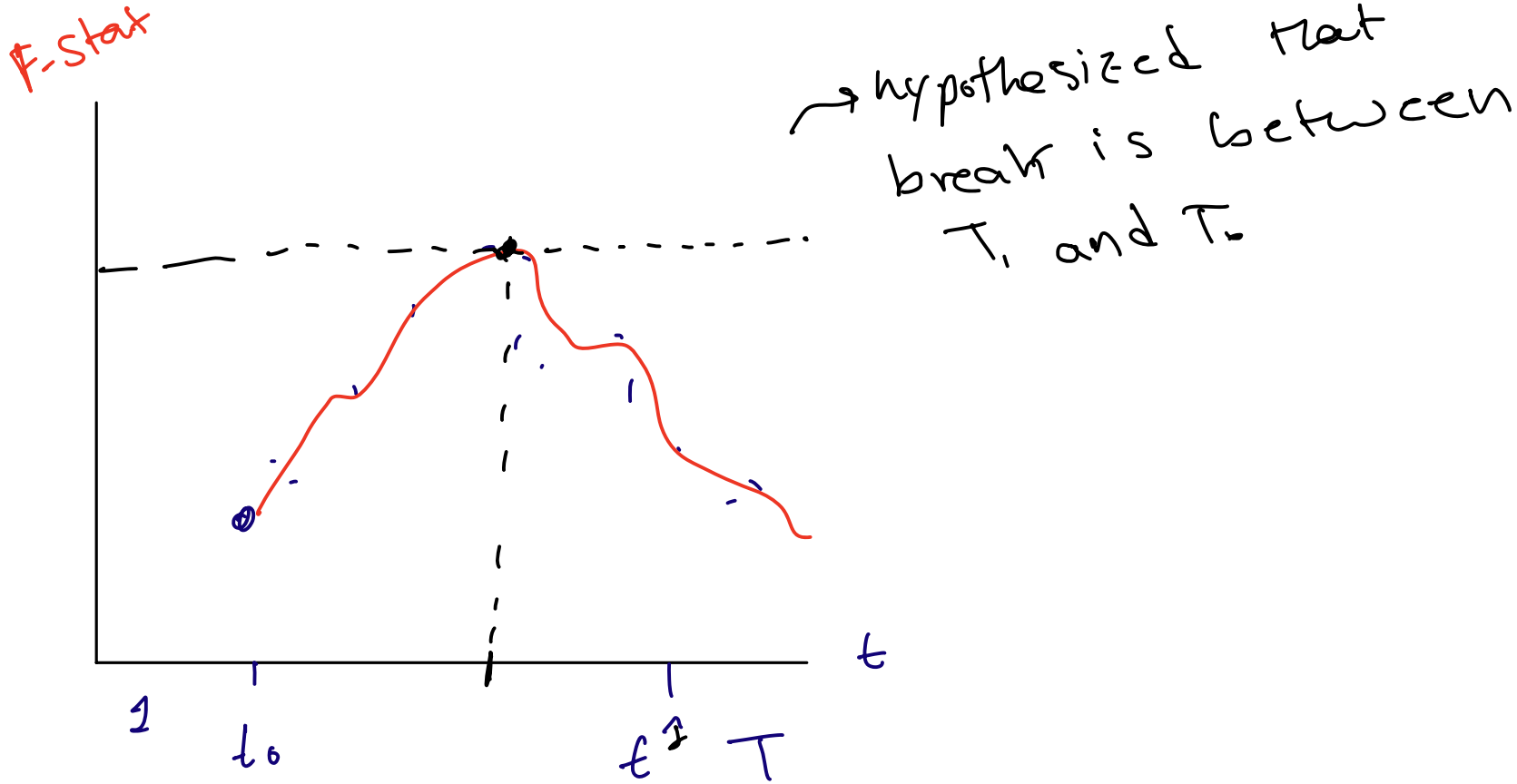
The Critical Value Problem

Here's the key insight the slide is making: you can't use standard F critical values.

Why? Because you're cherry-picking the largest F-statistic out of many tests. Even if there's no break, some date will have the highest F just by chance. This is a multiple testing problem on steroids.

The distribution of the supremum of F-statistics under the null (no break anywhere) is non-standard. Andrews (1993) and others have tabulated these critical values—they're considerably larger than regular F critical values.

The break is not in the mean, it is in the variance



Most likely candidate for break? highest point

The Graph

The red curve shows F-statistics at each possible break date. The test asks: is the peak of this curve high enough to reject "no break"? The annotation suggests the break is hypothesized to be between T_1 and T_2, and you're looking for the maximum within that range.

Practical Note

If you reject the null, the date where the F-statistic is maximized is your estimated break date—but the uncertainty around that estimate is often substantial.

Digression: Use of Dummy Variables

Dummy variables are very useful. Once you understand the role of the dummy variables and the interactive terms, we can apply the same principle to many questions:

- Test if the demand is on average different at different times of the years (i.e. seasonality, more examples later).
- Test if the elasticity of demand is different at different times of the year (car sales example etc).
- Isolate specific times of the year

Remember the question before. How many data points do we use in each of the periods we isolate?

Seasonality

Seasonality

- Seasonality is a pattern that repeats every year.
- From a micro point of view, seasonality comes from links of technology, preferences and institutions to the calendar. (examples?).
- We will only look at deterministic seasonality = repetition is exact and predictable.
- Seasonality is a very typical component of Time Series.

How do we deal with Seasonality?

Two choices: seasonally / non-seasonally adjusted

1. If we are interested in forecasting non seasonal fluctuations we may want to remove seasonality and work with seasonally adjusted series.

   Do we really want to do this? In general we want to forecast all variations in the series.

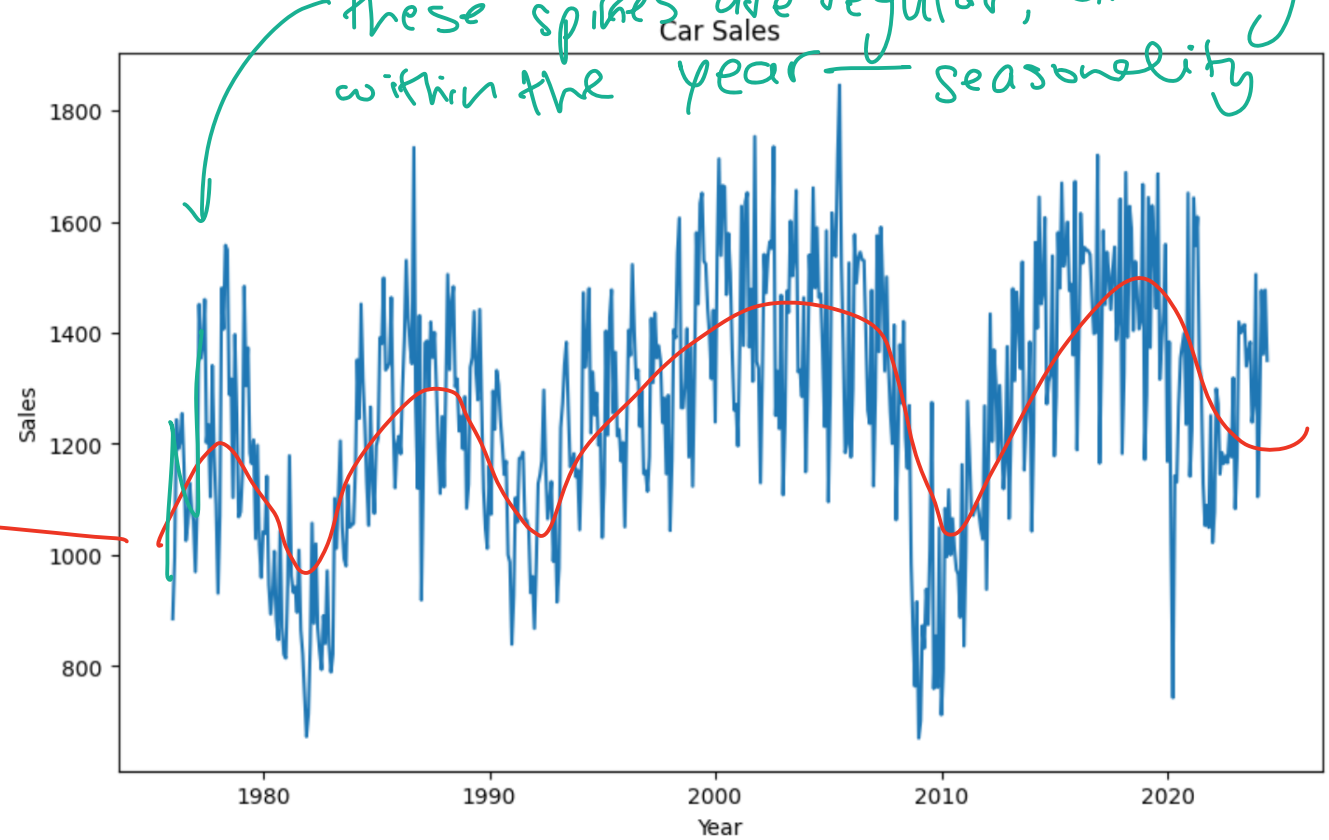
2. We can take seasonality into account in our forecast and model seasonality.

   Often data are already seasonally adjusted (SA), be careful when you download data which kind of data you really want. Data that is SA is passed through a complicated filter.

How to Model Seasonality

- To model seasonality, we use dummy variables!
- To motivate, let us use the cars.csv dataset which spans from 1967:1 to 2024:6
- It looks like the sales of cars are very different.

these spikes are regular, and they are within the year — seasonality



cycles

goes down in beginning of decades?

sales, cars, liquor — seasonality

4 seasons –

D_1 D_2 D_3 D_4

1967 : 1

2

3

4

1968 : 1

2

3

4

1969 : 1

2

3

4

Reminder: Dummy Variable Trap

- If you run your regression with the constant, you need to define one less dummy than categories (in the example, 4 seasons $\implies$ 3 dummy variables).
- Alternatively, you can define the same number of dummies as categories and then run the regression without the constant.

The problem: If you have 4 seasons and include 4 dummies (D_1, D_2, D_3, D_4) plus a constant, you have perfect multicollinearity. Why? Because $D_1 + D_2 + D_3 + D_4 = 1$ for every observation—it's identical to the constant term.
Solution 1: Include constant, use k-1 dummies (3 seasonal dummies). One season becomes the "reference category."
Solution 2: Include all k dummies, no constant.
Both work; they just give different coefficient interpretations.

Intercept and Slope in a Dummy Model

We formulate a pure seasonal dummy model w/ one explanatory variable (for now)

* Q_4 is omitted category

$$cars_t = c + \beta_1 Q_1 + \beta_2 Q_2 + \beta_3 Q_3 + \beta_4 price_t + u_t$$

Clearly from here, we know that

$$\mathbb{E}(cars_t | winter) = c + \beta_1 + \beta_4 price$$

$$\mathbb{E}(cars_t | spring) = c + \beta_2 + \beta_4 price$$

$$\mathbb{E}(cars_t | summer) = c + \beta_3 + \beta_4 price$$

$$\mathbb{E}(cars_t | fall) = c + \beta_4 price$$

The intercept is different for each season. What does this mean in economic terms?

Some Results

Model Diagnostics
$R^2 = 0.191$: The seasonal dummies and price only explain ~19% of sales variation. Lots of other factors matter.
Durbin-Watson = 0.858: Still suggests positive serial correlation—sales in adjacent quarters are correlated beyond what seasonality captures. Might want AR terms here too.

```
                            OLS Regression Results
==============================================================================
Dep. Variable:              Car Sales   R-squared:                       0.191
Model:                            OLS   Adj. R-squared:                  0.185
Method:                 Least Squares   F-statistic:                     34.00
Date:                Tue, 20 Aug 2024   Prob (F-statistic):           1.69e-25
Time:                        15:28:13   Log-Likelihood:                -3906.6
No. Observations:                 582   AIC:                             7823.
Df Residuals:                     577   BIC:                             7845.
Df Model:                           4
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Constant     826.2518     46.051     17.942      0.000     735.804     916.700
Q1           -11.4374     23.436     -0.488      0.626     -57.467      34.592
Q2           140.0223     23.435      5.975      0.000      93.995     186.050
Q3            69.5385     23.556      2.952      0.003      23.272     115.805
Car Price      2.9879      0.330      9.048      0.000       2.339       3.637
==============================================================================
Omnibus:                        2.671   Durbin-Watson:                   0.858
Prob(Omnibus):                  0.263   Jarque-Bera (JB):                2.658
Skew:                          -0.165   Prob(JB):                        0.265
Kurtosis:                       2.973   Cond. No.                         768.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

very low

statistically not significant compared to Q_4,

For a given car price, the average sales are higher in the second and third quarters than in the fourth quarter.

Now we can see the seasonal pattern quantified. Let me walk through the interpretation.

The Seasonal Coefficients (Relative to Q4)

Quarter	Coefficient	p-value	Interpretation
Q1 (Winter)	-11.44	0.626	~11 fewer sales than Q4, but **not significant**
Q2 (Spring)	140.02	0.000	~140 **more** sales than Q4, highly significant
Q3 (Summer)	69.54	0.003	~70 **more** sales than Q4, significant

Q4 (Fall) is the reference category—its effect is absorbed into the constant (826.25).

Economic Interpretation

The slide's conclusion is right: holding price constant, sales are significantly higher in Q2 and Q3 than in Q4. Q1 isn't statistically different from Q4.

This makes intuitive sense for car sales:

- Spring (Q2): Tax refunds arrive, people buy cars
- Summer (Q3): Good weather, family road trips, new model year anticipation
- Winter (Q1) and Fall (Q4): Lower demand

The Price Effect

Car Price coefficient = 2.99 ($p < 0.001$): This seems counterintuitive—higher prices associated with more sales? This is likely picking up an endogeneity issue or reverse causality. When demand is high, prices rise. You'd need an instrument to estimate the true causal effect of price on quantity demanded.

Some Extensions

The seasonal dummies capture systematic within-year variation in car demand. Q2 and Q3 being higher reflects consumer behavior patterns—tax refunds in spring, summer vacation planning, new model years arriving in fall (so people buy the "old" models at discount in late summer).
The positive price coefficient is the puzzle. Economically, demand curves slope downward—higher prices should mean fewer sales. The positive estimate likely reflects simultaneity bias: we're observing equilibrium outcomes where high demand drives both higher sales and higher prices. To get the true price elasticity, you'd need an instrument that shifts supply but not demand.

- What is the economic intuition of our estimates?
- What other variable/s would you include in this model?
- Do you think that differences in the average demand for cars across seasons is the only kind of seasonality we need to take into account?

Is a Level Shift the Only Seasonality?
No—and this is a great question. The current model assumes the intercept shifts by season, but the slope (price sensitivity) stays constant. But consider:

- Maybe consumers are more price-sensitive in Q4 when demand is low
- Maybe the price effect differs in recession years

This calls for interaction terms: $Q_1 \times$Price, $Q_2 \times$Price, etc. That would allow the price elasticity to vary by season—a much richer model.

Interaction Terms

- It may also be that not only is the sales higher in Q2 for a given price but also that the demand for cars is more sensitive to the interest rate in the different seasons!
- We want to model possible shifts in the slope of the regression line for different seasons
- We can do this by looking at the interaction of the dummy variables with the price variable

Interactive Dummy Variables

Consider this simple model with intercept and slope dummy

$$cars_t = c + \beta_1 Q1 + \beta_2 Q_2 + \beta_3 Q_3 + \beta_4 p_t + \beta_5 (Q_1 \cdot p_t) + \beta_6 (Q_2 \cdot p_t) + \beta_7 (Q_3 \cdot p_t) + u_t$$

```
                            OLS Regression Results
==============================================================================
Dep. Variable:              Car Sales   R-squared:                       0.195
Model:                            OLS   Adj. R-squared:                  0.185
Method:                 Least Squares   F-statistic:                     19.84
Date:                Tue, 20 Aug 2024   Prob (F-statistic):           7.26e-24
Time:                        16:13:23   Log-Likelihood:                -3905.1
No. Observations:                 582   AIC:                             7826.
Df Residuals:                     574   BIC:                             7861.
Df Model:                           7
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Constant     748.0431     89.883      8.322      0.000     571.503     924.583
Q1           142.5440    123.148      1.157      0.248     -99.332     384.420
Q2           279.7676    124.202      2.253      0.025      35.822     523.713
Q3            72.6679    126.668      0.574      0.566    -176.122     321.457
Car Price      3.5895      0.679      5.283      0.000       2.255       4.924
Q1_Price      -1.1878      0.932     -1.274      0.203      -3.019       0.644
Q2_Price      -1.0761      0.939     -1.146      0.252      -2.921       0.769
Q3_Price      -0.0206      0.960     -0.021      0.983      -1.907       1.865
==============================================================================
Omnibus:                        2.873   Durbin-Watson:                   0.850
Prob(Omnibus):                  0.238   Jarque-Bera (JB):                2.879
...
Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

stayed the same

coefficients shot up

was significant before, no longer significant

What is the underlying model for each season?

Your Annotations Capture the Key Findings

"R^2 stayed the same": Adding the interaction terms didn't improve fit much—the extra flexibility wasn't needed.

"Coefficients shot up": The seasonal intercept coefficients (Q1, Q2, Q3) got larger, but look at the standard errors—they also got larger. This is because the interactions are correlated with the level dummies, inflating standard errors.

"Was significant before, no longer significant": This is the key insight. Q1 was already insignificant, but now even Q2 and Q3 interactions with price (Q2_Price, Q3_Price) aren't significant. The price sensitivity doesn't appear to vary meaningfully across seasons.

The Lesson

More flexible isn't always better. The interaction terms added complexity without improving the model—the simpler model with just seasonal intercepts was sufficient. This is where AIC/BIC would help you choose.

Joint F-test question: Even if individual interactions aren't significant, would you want to test them jointly before dropping them?

Forecasting with Seasonality

- Given that we are only looking at deterministic seasonality, forecasting with seasonal dummy variables is very easy! (i.e. we know exactly how the dummy variable looks in the future)
- You can predict what the value for the dummy will be next quarter and you can use the estimated values for your parameters to forecast.
- We can re-estimate using only part of our sample and see how we are doing in forecasting.
- We don't have a good R^2 so most likely we will not do too well in forecasting.